IMO 2026 — Problem 2

Solution by GPT-5.6 Sol

2026-07-18

Source: `results/gpt-5.6-sol/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle LBA, L lies inside the angle ACK, and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Solution document

Status

solved

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Approaches tried

- Trigonometric/complex coordinates: successful. Coordinates of K, L follow from the sine rule in BMK, CNL; the two remaining angle conditions become reality relations.
- Equal powers: successful. A standalone complex-algebra lemma (`lemmas/complex-algebra.md`) shows that those reality relations imply that M, N have equal powers with respect to (AKL).
- Symbolic Laurent-polynomial checks (`code/symbolic.py`, `code/eliminate.py`, `code/certificate.py`) were used to check the algebra and orientations. An apparent orientation correction was rechecked: for counterclockwise ABC, the ray directions are $CL = \alpha + \pi + x$ and $CK = \alpha + \pi + x + z$.

Current best

The claim follows by normalizing $A = 0, B = 2, C = 2de^{i\alpha}$. Then $M = 1, N = de^{i\alpha}$, while the sine rule gives explicit complex coordinates for K, L. The angle order assumptions turn the remaining conditions into exactly the two hypotheses of the complex equal-power lemma, proving that M, N have equal powers with respect to (AKL). As both powers equal the squared distance to O minus the same circumradius squared, $OM = ON$.

Full proof

Reflect the whole configuration if necessary, and assume that A, B, C occur counterclockwise. We use complex coordinates, normalized by

$$A = 0, \qquad B = 2, \qquad C = 2n, \qquad n = de^{i\alpha},$$

where $d > 0$ and $\alpha = \angle BAC$. Thus

$$M = 1, \qquad N = n.$$

Set

$$x = \angle KBA = \angle ACL, \quad y = \angle LBK = \angle LNC, \quad z = \angle LCK = \angle BMK,$$

and abbreviate

$$X = e^{ix}, \qquad Y = e^{iy}, \qquad Z = e^{iz}, \qquad a = e^{i\alpha}.$$

We first find K. In triangle BMK we have

$$BM = 1, \qquad \angle MBK = x, \qquad \angle BMK = z.$$

Consequently $\angle BKM = \pi - x - z$, so the sine rule gives

$$MK = \frac{\sin x}{\sin(x+z)}.$$

The ray MK makes angle z with the positive real axis; hence

$$K = 1 + \frac{\sin x}{\sin(x+z)} Z. \qquad (1)$$

Likewise, in triangle CNL we have $CN = d$, $\angle NCL = x$, and $\angle CNL = y$. The ray NL has direction $\alpha - y$, and the sine rule gives

$$L = n\left(1 + \frac{\sin x}{\sin(x+y)} Y^{-1}\right). \qquad (2)$$

The denominators in (1)–(2) are nonzero because they are sines of angles supplementary to angles of the nondegenerate triangles BMK, CNL.

We next translate the ordering of the rays. Since K lies inside $\angle LBA$, when one turns inside that angle from BA toward BL, one meets BK first. Therefore BL has direction $\pi - x - y$. It follows that

$$(L-2)XY \in \mathbb{R}. \qquad (3)$$

At C, the ray CA has direction $\alpha+\pi$. Since $\angle ACL = x$, the ray CL has direction $\alpha+\pi+x$. Since L lies inside $\angle ACK$ and $\angle LCK = z$, the ray CK has direction $\alpha+\pi+x+z$. Thus the opposite vector $C-K = 2n-K$ has direction $\alpha+x+z$, and hence

$$\frac{2n-K}{aXZ} \in \mathbb{R}. \qquad (4)$$

We now apply the complex equal-power lemma proved in `lemmas/complex-algebra.md`; its short algebraic proof is reproduced here for self-containment. If p, q denote the complex coordinates of K, L, respectively, then, since

$$\frac{\sin x}{\sin(x+z)} = \frac{X-X^{-1}}{XZ-X^{-1}Z^{-1}}, \qquad \frac{\sin x}{\sin(x+y)} = \frac{X-X^{-1}}{XY-X^{-1}Y^{-1}},$$

formulas (1)–(2), together with (3)–(4), satisfy the hypotheses of that lemma. We recall its verification.

Write $t^* = \bar{t}$. The circle through $0, p, q$ has equation

$$w\bar{w} + uw + v\bar{w} = 0. \qquad (5)$$

Putting $w = p, q$ and applying Cramer's rule gives, with $\Delta = pq^* - qp^* \neq 0$,

$$u\Delta = -pp^*q^* + qq^*p^*, \qquad v\Delta = -pqq^* + qpp^*. \qquad (6)$$

The difference between the values of the left side of (5) at 1 and at n is, after multiplication by Δ,

$$\begin{aligned} H = (1-nn^*)\Delta + (1-n)(-pp^*q^* + qq^*p^*) \\ + (1-n^*)(-pqq^* + qpp^*). \qquad (7) \end{aligned}$$

The two reality relations (3)–(4) are equivalent to

$$(q-2)XY = (q^*-2)X^{-1}Y^{-1}, \qquad (8)$$

$$(2n-p)a^{-1}X^{-1}Z^{-1} = (2n^*-p^*)aXZ. \qquad (9)$$

Substitute

$$p = 1 + Z\frac{X-X^{-1}}{XZ-X^{-1}Z^{-1}}, \qquad q = da\left(1 + Y^{-1}\frac{X-X^{-1}}{XY-X^{-1}Y^{-1}}\right)$$

and their conjugates into (7). On using (8)–(9), and putting terms over the common denominator

$$(XZ-X^{-1}Z^{-1})^2(XY-X^{-1}Y^{-1})^2,$$

the numerator cancels in conjugate pairs: the four types are

$$(nq^*p^*, -n^*qp), \quad (npp^*q^*, -n^*pp^*q), \quad (qq^*p^*, -qq^*p), \quad (pq^*, -qp^*).$$

Thus $H = 0$. By (5)–(7), the powers of $1 = M$ and $n = N$ with respect to the circle through $0, p, q$, namely (AKL), are equal.

Let R be the radius of (AKL). The power of any point P with respect to this circle is $OP^2 - R^2$. Therefore

$$OM^2 - R^2 = ON^2 - R^2,$$

so $OM^2 = ON^2$. Both are nonnegative lengths, and hence

$$\boxed{OM = ON}.$$